

Below you'll find the instructions for the final exercise. **Please Submit it by 31.07.26.**

Submission will be in pairs unless you receive explicit approval to submit on your own.

The final assignment in the course consists of two modules.

A short task practicing ML methods -

Please load the [Fashion MNIST dataset](#). Using the existing training/test split. Attempt to achieve an accuracy score of at least 0.8 on the test set. Here, try to use SVM and Ensemble methods.

Report on your model's results and comment on its generalizability on new data sets.

Then, attempt to run PCA to compress the dataset. Visualize the numbers with the original data and the compressed data. How many PCs are needed to obtain a compressed data set such that the visualized numbers are clear?

Main assignment -

(a) Select a dataset which requires EDA and preprocessing (i.e., not a fully clean dataset) and prepare it for analysis.

The dataset can be your own research data, or a dataset from the internet (see options from these repositories - [1](#), [2](#), [3](#)). The dataset can also be from a research paper which provided the data.

Here are more considerations/limitations -

1. Generally, text datasets aren't an option as we didn't cover NLP. You can choose either tabular or image datasets.
2. Try to avoid nested/dependent (e.g., longitudinal, time series) data, or choose a model which does not make assumptions regarding the independence of observations. Make sure to avoid data leakage between train and test datasets, if you are using data with nested observations. Alternatively process the data that it would not contain dependent observations (e.g., aggregation).
3. Don't select a very simple ("toy") dataset (e.g., Iris), but rather one that is more challenging (e.g., the [Kaggle housing dataset](#)). On the other hand, don't select an overly complex dataset.
4. Please let me know if you can't find a dataset, after you have put some time into it.

(b) Analyze the dataset by applying to the dataset both a supervised learning technique (classification\ regression), and an unsupervised learning technique (clustering\ dimensionality-reduction).

1. Discuss the performance of both models in terms of complexity, performance and generalization. Don't use neural networks, but "classical" machine learning algorithms (you can however use ones which we haven't covered in the course, e.g., LDA). The classification/regression estimator you are using should be compared to a simpler estimator trained on the same task (e.g., fewer features, more regularized). Whether you are required to do classification **or** regression depends only on the data you select.

2. The application of the unsupervised technique should be justified (e.g., reducing excess features, using clustering to collapse similar categories which have a low proportion of cases, outlier detection, etc.).

(c). Misc.

1. Your assignment should be in the form of a Jupyter notebook, with separate code/markdown cells, appropriate headers and logic.
2. All cells should be executable, in the order they appear in the notebook. Don't leave large portions of commented out code.
3. Make the analysis process explicit by specifying what you are trying to accomplish at each step (in the preceding markdown cell).
4. As a rule of thumb, it should take an evaluator 10-15 minutes to go over your submission, including reading the comments and executing the code.
5. If your model is included in an extension of Sklearn – the notebook should include cells referring to the installation of this package, and it should be runnable on Google Colab.
6. Your notebook should include loading the data (e.g., using URL for a site or a shared folder, for example).
7. If training your model takes more than 3-4 minutes, you should look into saving it (see [here](#)), so that the model can be loaded when running your submission.